DATA SET DS1507

16. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1 – 10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8 −	12	8	5 +
2	4	5 −	13	4	2 +
3	7	4 +	14	3	3
4	8	7 +	15	6	8 −
5	2	3 −	16	9	10 −
6	7	4 +	17	9	8 +
7	9	9	18	4	6 −
8	7	8 −	19	4	3 +
9	2	5 −	20	5	4 +
10	4	3 +	21	3	2 +
11	6	9 −	22	5	3 +

a. Use the binomial tables in Appendix I to find the exact rejection region for the test.

b. Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)

c. Compare the results of parts a and b.

$H_0: P(A>B) = p = 0.5$

$H_a: P(A>B) \neq p \neq 0.5$

$n = 20,\ \alpha = 0.05$

$(+) = 11 \quad (-) = 9$

$$P(X \geq 11) = P(X=20) + P(X=19) + P(X=18) + P(X=17) + P(X=16) + P(X=15) + P(X=14) + P(X=13) + P(X=12) + P(X=11)$$

$$= 0.412$$

$p = 2 \times 0.412 = 0.824 > 0.05$

$\Rightarrow$ non-reject H_0

There is no sufficient evidence to say that one of the chefs tends to give higher rating than other.

b. $\alpha = 0.05$

$$z_{STAT} = \frac{x - 0.5n}{0.5\sqrt{n}} \qquad z^* = \frac{11 - 0.5(20)}{0.5\sqrt{20}} = 0.4472$$

RR: $|z| \geq 1.96$ $\quad 0.4472 < 1.96 \Rightarrow$ non-reject H_0.

There is no sufficient evidence to say that one of the chefs tends to give higher rating than other.

c. The result are same.